

KILLA COSMETICS

Project Completion & Refactoring Report

Date: May 24, 2026

Developed by: Jules (AI Agent)

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1. Executive Summary

This report details the comprehensive audit, refactoring, and completion of the Killa Cosmetics web application. The project, originally a procedural PHP application, has been fully transitioned to a robust MVC (Model-View-Controller) architecture. All core functionalities promised in the README have been implemented, secured, and verified. The result is a professional, functional, and production-ready e-commerce management system.

2. List of Changes

Modified Files:

- src/Config/Database.php: Updated to use environment variables for better security.
- src/Core/Controller.php: Added support for layouts, RBAC, and JSON responses.
- src/Core/Router.php: Implemented a robust routing system.
- src/Models/Product.php: Added pagination, advanced search, and count methods.
- src/Models/Statistics.php: Fixed queries and table references.
- src/Services/ExportService.php: Fully implemented PDF, Excel, and CSV export logic.
- src/Views/dashboard/inventory.php: Updated to dynamic data and integrated with layout.
- basekilla2.sql: Added missing table 'detalle_ventas'.

New Files:

- public/index.php: Created as the main entry point (Front Controller).
- src/Controllers/AuthController.php: Handled login, logout, and session management.
- src/Controllers/DashboardController.php: Managed administrative dashboard logic.
- src/Views/auth/login.php: Login interface.
- src/Views/products/list.php: Product listing with search and pagination.
- src/Views/products/form.php: Unified form for creating and editing products.
- src/Views/layouts/main.php: Master template for consistent UI.
- .env / .env.example: Environment configuration management.
- composer.json: Dependency management.

3. Security Improvements

- Prepared Statements: All database interactions use mysqli prepared statements to eliminate SQL injection risks.
- Password Hashing: Integrated password_hash() for storage and password_verify() for authentication. Plaintext passwords were converted via update_passwords.php.
- RBAC: Implemented Role-Based Access Control to ensure users only access functionalities permitted by their role (Admin, Cashier, etc.).
- Input Sanitization: Used a dedicated Validator class to clean and validate all user inputs (POST/GET).
- Secure Env: Sensitive credentials moved from code to .env files.

4. Architecture Changes

- Procedural to MVC: Moved away from scattered .php files to a centralized Controller-Model-View structure.
- Front Controller Pattern: All requests flow through public/index.php, improving security and maintainability.
- PSR-4 Autoloading: Integrated Composer for clean class loading and dependency management.
- Service Layer: Complex business logic like data export isolated in the Services directory.

5. Features Added

- Full CRUD: Complete management of products including image uploads.
- Search & Pagination: High-performance product browsing with server-side pagination.
- Dashboard: Real-time statistics including low stock alerts and top selling products.
- Multi-format Export: Professional exports to PDF, Excel, and CSV.

6. Bugs Found and Fixed

- Missing Table: The 'detalle_ventas' table referenced in statistics was missing from the SQL dump; it has been added.
- Path Errors: Fixed broken relative paths for includes and assets.
- Incomplete Logic: Finalized placeholder methods in ExportService and Statistics model.
- Variable Conflicts: Resolved undefined variable notices across views.

7. Testing Performed and Results

- Auth Flow: Verified login/logout and session persistence across roles.
- CRUD: Tested creation, update, and deletion of products.
- RBAC: Confirmed restricted access for non-admin users.
- Exports: Verified generated PDF, Excel, and CSV files.

8. Pending Improvements

- Categories Management: Implement a dedicated CRUD for categories.
- Shopping Cart: Add frontend shopping cart functionality for Client role.
- Advanced Analytics: Add charts to the dashboard.